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Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection

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Abstract Infected Grouper, *Epinephelus tauvina* were collected from fish landing center Kadiappatinam of Kanyakumari district, South India and found out the distribution of different bacterial genera from body fluid and blood. Bacterial genera such as *Vibrio* sp. (29 %), *Streptococcus* (29 %), *Pseudomonas* (14 %), *Cornybacterium* (14 %) and *Bacillus* (14 %) from tissue samples and *Vibrio* sp. (40%), *Staphylococus* (20 %), Proteus (20%) and *Bacillus* (20 %) from blood samples were detected. Herbals such as *Aegle marmelos, Terminalia bellirica, Solanum trilobatum, Adathoda vasica* and *Leucus aspera were* extracted with butanol, petroleum ether, acetone and benzene and tested against the pathogen *Vibrio harveyi* isolated from the infected Grouper, *E. tauvina*. Among the different solvent extractions to *V. harveyi*, petroleum ether extracts were suppressed significantly (P<0.05). The equal proportion of the all plant extracts were mixed with the artificial feeds at the concentration of 100, 200, 400 and 800 mg/kg of diet and fed the diets to the grouper juveniles from 20 ± 2 g average weight for a period of 60 days. On every 20 days fishes were challenged with *V. harveyi* and survival, growth parameter, immune response was studied. The herbal diets significantly (P< 0.05) increased the survival. Surprisingly after 60th day challenge, the herbal diets yielded very good results and attained the survival of 75.35, 82.0, 89.0 and 92.0 % in the concentration of 100, 200, 400 and 800 mg/kg, respectively. Like survival, the average wt gain (g/day) significantly increased (P<0.05) from the control diet fed fish (0.19 g/day) to 400 mg/kg diet (0.53 g/day). Also significant improvements (P<0.05) in immune parameters such as phagocytic activity and Albumin-Globulin (A-G) ratio were observed with the fish fed on different concentrations of herbal diets compared to the control group. The phagocytic activity and Albumin-Globulin (A-G) ratio were increased two times in control diet fed fish and increased more than one times in experimental diet fed fishes. Among the different concentrations of the herbals in the diet, 400 mg/kg was the most effective concentration and is recommended for grouper disease management.

Key Words Antibacterial; Herbals; *Epinephelus tauvina*; Immunostimulants; *Vibrio harveyi*

Introduction

Since market demand for groupers is larger than supply from captures fisheries and considering the fast growing economic importance of grouper culture in recent years, nowadays many countries are concentrating on developing commercial grouper farming technology (Al-Thobaity and James, 1996). Many commercial grouper fish farms depend mainly on juveniles collected from the wild due to hatchery constraints and poor larval survival, that limits commercial expansion of grouper farming in south-east Asian countries (Chen, 1979; Tseng, 1983; Quinito and Toledo, 1993). In Hong Kong, the grouper production was about 3000 t/year on early 1990's. In the last few years, production has

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dropped to 1000 t/year due to environmental problems such as poor water conditions leading to microbial infections and disease outbreaks (Sadovy, 2000). A number of diseases for groupers have been reported and the main causative agents are the typical group of parasitic organisms, bacteria and viruses. Pathogenic bacteria cause severe losses in hatchery and grow-out culture systems which are usually related to poor management and water quality. Fish are easily stressed by improper handling and can become infected with *Flexibacteria*, leading to skin damage and tail rot. The gram negative bacteria, especially *Vibrio parahaemolyticus* and *V. alginolyticus* were cause haemorrhagic septicaemia in grouper at nursery and grow-out cage systems. The gram-positive bacteria *Streptococcus sp.* also causes systemic infection in grouper. Affected fish exhibit signs of weakness, abnormal swimming and occasionally corneal hemorrhage (Danayadol *et al*., 1995). Several bacterial diseases has also been reported, including *Vibrio* spp., *Pseudomonas* sp, *Pasteurella piscicida* and *Flexibacter* sp. Bacteriosis caused by *Pseudomonas* sp. among *E. tauvina* cultured in Malaysia was first reported by Nash *et al*. (1987), where all the age groups were affected during an outbreak in November/December to February/March in 1982-1986. Mortality was ranged from 20% to 60%. Affected fish showed extensive haemorrhagic erosions and ulcerations of the skin, fins and tail.

Antibiotic and other chemical treatment is widely used to control infectious parasitic, bacterial and fungal diseases. It has been used for control of parasitic and bacterial diseases of grouper. However, earlier study demonstrated that there are problems in chemical treatment of fish diseases; in some cases the treatments can be harmful because of associated stress (Chang and Chao, 1986). In Singapore, Chong and Chao (1986) reported some cases of drug overdose leading to fish death and other detrimental side effects. For example, formalin overdose results in severe gill damage. Ulcerative dermatitis develops with repeated treatment with nitrofurazone leading to death of fishes. Application of chemotherapeutants has created problems with toxicity, resistance, residues and possibly some public health and environmental consequences. Their efficacy under certain aquatic conditions (open water systems) remains questionable, and is expensive when compared with herbal application. The use of chemicals in treating health problems has also been complicated by the misleading advice provided to the farmers by feed and chemical companies regarding the use of antibiotics and other therapeutic drugs.

Active principles of plant origin, which were isolated, promise to deal with drug resistant bacteria (Richard *et al*., 1983). Compounds such as volatile oils, tannins, phenolic compounds, saponins, alkaloids polysaccharides and polypeptides were shown effective alternatives to that of antibiotics. The screening of plant extracts and natural products for antimicrobial activity has shown that higher plants represent a potential source of new anti-infective agents (Press, 1996) as well as serve in drug discovery from natural products for primary lead compounds (Lawrence, 1999). Antibacterial therapy using chemicals is going through a crisis due to rapidly increasing development of resistance to existing agents. Such resistance has an impact on all areas of chemotheraphy. Plant derived phytomedicines has great promise in the treatment of infectious diseases. Plant based antimicrobials represent a vast untapped source for medicines and has enormous

therapeutic potential. They are effective in the treatment of infectious diseases while simultaneously mitigating many of the side effects that are often associated with synthetic antimicrobials.

Herbals having the characteristics of antimicrobial, anti stress, immunostimulant, growth promoting plant products significantly influenced the fish / shrimp larviculture (Citarasu et al., 1998; Citarasu et al., 2002; Citarasu et al., 2003a; Citarasu et al., 2003b). Sivaram et al. (2004) successfully controlled the Vibrio pathogen, and improved the immune system of the grouper larviculture using herbal methonolic extracts. These findings suggested that phytochemicals could be an alternative to the chemotherapeutic molecules and safe to use in aquaculture. The present study focus on the isolation and characterization of the bacterial pathogens from the infected grouper and screening the herbals against selected pathogens and apply the products to Grouper larviculture.

Materials and methods

Collection of samples

The infected experimental fish *Epinephelus tauvina* (56 cm total length and 20.8 kg wt) were collected from fish landing center at Kadiappatinam, Kanyakumari district, south India. The collected samples were aseptically taken into the laboratory and performed the bacteriological examinations. Then the samples were stored in deep freezers (-20) for further experiments to keep the sample from microbial destruction and biochemical changes such as rancidity etc.

Isolation and identification of bacterial flora from infected grouper E. tauvina

Three samples were collected from ulceration (wounds) of diseased fish aseptically and blended with Alkaline Peptone Water (Hi media) with 3% NaCl (APW 3%). Body fluid was collected using sterile hypodermal syringe. Source of bacterial samples using mucus swabs collection on the body surface of the fish and was also done aseptically. All these samples were transferred separately to APW 3% in 500 ml conical flasks and incubated at 37°C as it is the optimum temperature for bacteria. One gram of muscle tissue samples was homogenized and serial dilutions were made at 10^2 to 10^7 concentration. One ml of each dilution was inoculated into Zobell Marine agar (Hi media) plates by both spread and pour plating methods and incubated at 37°C for 24 hours. Likewise, 1 ml blood taken from the experimental fish tissues was taken and serial dilutions were made at 10^2 to 10^6 concentration. Plates with 50 – 300 colonies were taken for enumeration. Using colony morphology, shape and colour, major dominant flora was determined. Dominant species were selected and again streaked into Marine Zobell agar plates, as it is a common media for most of the marine bacteria. Pure cultures were selected from these plates and slants were made. Further conformation was done according to standard procedures described in Bergey's manual of determinative bacteriology (Table 1).

Selected herbals

Herbals having that antimicrobial and immunostimulant characteristics were selected following Nadkarni (1995).

Organic extraction and screening of herbals against the dominant pathogen Vibrio harveyi

Herbal powders were soaked with butanol, petroleum ether, benzene and acetone

individually for 48 h (Eloff, 1998) in equal ratio (1:1). The slurry was then filtered, washed to remove non–soluble fractions and filtrate was centrifuged (20,000 X g for 30 min) for clarification. The clarified extracts were condensed at 35°C until the solvent residue had evaporated. Each 500 µg of condensate was impregnated in 5 mm diameter sterile paper discs (Himedia, India) and screened against *V. harveyi* through replicates of disc diffusion test as described in Bauer *et al* (1966). Dried antimicrobial discs with impregnated herbal extracts were carefully dispensed with uniform distances over agar surface and ensured for correct implantation by applying gentle press over disc. All plates were kept in 35°C for 24 hours incubation as it is the optimum temperature for the bacteria. After the incubation time, plates were examined for inhibitory zone formation of antimicrobial extracts.

Minimum inhibitory concentration (MIC)

The MIC was defined as the lowest concentration of a fraction of antimicrobial extract, which was able to completely inhibit the growth of each bacterial strain. Extracted materials were dissolved in 10% Dimethyl sulphoxide (DMSO) and added to Muller Hinton agar broth (pH 7). Fifty microlitre of liquid bacterial culture was added into each tube, which had been added already with different concentration of extracts (10 µg – 100 µg). The MIC values were taken as the lowest concentration of extract that inhibited the growth of the organism after 24 hrs of incubation at 37°C (Jacobs and Demott, 1994).

Determination of Minimum Bacterial Concentration (MBC)

The Minimum Bacterial Concentration (MBC) (Jacobs and Demott, 1994) was determined by plating the culture from the tube without the growth of microorganisms. To determine the minimum bacterial concentration, aliquots of one loopful of the two lowest concentrations, those inhibited bacterial growth were streaked on petri plates containing Muller Hinton agar. After an overnight incubation at 37°C, the plates were evaluated by comparing them with control plates containing bacteria without test compounds. The lowest concentration that gave no visible growth was taken as MBC.

Experimental fish culture

Juvenile of *E. tauvina* weighing approximately 20 ± 2g were caught from the Gulf of Mannar, India. They were acclimatized and kept in the quarantine tanks for a period of 15 days to assess their disease free health status. During that time, they were fed with commercial feeds. After acclimatizing, five triplicated (5x3=15) experimental groups were stocked having an average body weight of 22 ± 1 g in the 500 l capacity, flow-through aquaria with the water flow rate of 1 l/min. The water quality parameters were maintained constantly (salinity 30 ± 0.5 ppt, temperature 28 ± 1.5°C, DO >6.0 mg/l, ammonia-nitrogen <0.22 mg/l).

Diet preparation and feeding

Petroleum ether extracts of five herbals were taken and incorporated at equal proportion and make five experimental diets containing 0 (Control), 100, 200, 400 and 800 mg/kg of extracts mixture. Diets were prepared following the basal ratio of Boonyaratpalin (1993). The basal diet contained 45.1% protein, 7.2 lipid, 14.6% ash, 7.1% moisture and 3% fibre. To prepare the feeds, ingredients were mixed thoroughly and 4% gelatin solution containing active principles in

appropriate concentration was added along with oil ingredients. Sufficient water was added and the pH was adjusted to 7±0.1. Then the mixture was cold extruded, cut into pellet, air dried and stored at room temperature. Fish were fed with 5% of the body weight, every day till the end of experiment. Weight gain was determined following the method of Ricker (1979).

Bacterial challenge

Once in every two weeks, the 20, 40 and 60th day, 15 fishes were challenged with the lethal dose of freshly isolated (2 days old) *V. harveyi* (2x10^{8}) (isolated from the infected *E. tauvina)* cultured and maintained using Tryptic soy broth and agar with 2% NaCl (W/V) intramuscularly, and transferred to a 250 l aquaria and observed the pathological signs for up to five days.

Immunological parameters

During every 20 days interval, after 24 h of final feeding, five randomly chosen fishes from each experiment and control groups were taken and anaesthetized with 50 mg MS-222/dm^{3} of water, and used for blood collection. Blood was collected from caudal vein, with 1 ml plastic syringe rinsed with anticoagulant, and a part of the blood was transferred immediately, added equal volume of 10 % tri sodium citrate and stored at 4°C. The remaining blood was kept in room temperature for 1 hour without anticoagulant to collect the serum and stored at – 40°C for further analysis. For phagocytosis assay, 10^{7} cells of formalin-killed *V. harveyi* were added to 0.1 ml of pooled blood samples in a sterile micro plate and incubated for 30 min at 25°C after thorough mixing in the well. After incubation, the blood–bacteria suspension was mixed gently and 50 µl of this suspension was smeared on three glass slides. After air-drying, the smears were fixed in 95% ethanol, redried and stained with May–Grunwald's Giemsa. The phagocytic cells and phagocytosed bacteria were counted (Park and Jeong, 1996). Serum bactericidal activity, experimental fish *E. tauvina* of ten number were injected with 0.1 ml/kg body weight with live (*V. harveyi* 10^{5} CFU ml/l) and blood samples were collected at 10 and 90 min. after injection. A hundred microlitre blood was serially diluted and plated in the TCBS medium (Hi media, India). Albumin-globulin ratio (A-G) of the sera was calculated following the method of Sahoo *et al* (1999). The serum samples were analysed for total protein following the dye binding method of Bradford, using bovine serum albumin as standard, for albumin by the bromocresol green method, and for globulin by subtracting the albumin value from the total protein value. Finally, the albumin–globulin ratio (A-G) was calculated.

Data analysis

Values were analyzed using one-way ANOVA following Zar (1994) to discover the significant difference at 5% ($P<0.05$) level.

Results

Isolation of bacterial genera

The quantity of bacterial pathogens was analyzed from the infected groupers and the results are given in the Fig.1a & b. In total, there were five different major microbial genera in the tissue samples; such as *Vibrio* sp. (29 %); *Streptococcus* (29%); *Pseudomonas* (14%); *Cornybacterium* (14%) and *Bacillus* (14%) were detected in the tissue. In the blood samples, *Vibrio* sp. (40%); *Staphylococus* (20%); *Proteus* (20%) and *Bacillus* (20%) were detected. Among the different pathogens, *Vibrio* was dominant in both samples and

among the genus *Vibrio*, *Vibrio harveyi* was most abundant (data not given).

Herbal extract screening against *V. harveyi*

Petroleum ether (PE) was the most effective solvent and followed by benzene, butanol and acetone among the different solvent extractions (Fig. 2), to control the pathogen *V. harveyi* isolated from the infected grouper. The zones of inhibitions for petroleum ether extracts were 12, 13, 14.5, 11 and 10.26 mm. for *Aegle marmelos*, *Terminalia bellirica*, *Solanum trilobatum*, *Adathoda vasica* and *Leucus aspera respectively*.

Minimum inhibitory concentration and minimum bactericidal concentration

Minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) were studied and the results are given in Table 2. The MIC was carried out all petroleum ether extracts of the herbals against the dominant bacteria *V. harveyi*. The herbal extracts, such as *A. marmelos*, *T. bellirica*, *S. trilobatum* were active against *V. harveyi* at the concentration of 70 µg/ml. The MIC values obtained for the other herbal extracts such as *A. vasica* and *L. aspera* were 80 µg/ml.

Bacterial challenge

The control group succumbed to death hundred percent mortality with in 2 days of 20, 40 and 60th day challenge. Survival was significantly increased (P<0.05) for the herbal immuno stimulant incorporated diets fed groups. A slight variation in survival was observed only after the 20th day challenge among different concentrations of herbal diet. Maximum survival of 32.66% was observed in 400 mg/kg diet. Surprisingly on 60th day after challenge, the herbal diets yielded a survival of 75.35, 82.0, 89.0 and 92.0 % in the concentration of 100, 200, 400 and 800 mg/kg, respectively (Fig. 3).

Growth studies

An average weight gain (g/d) was calculated from the difference between the final and initial weight of each fish group (Table 3). The fish in the control group gained 0.19 g/d. This was significantly increased (P<0.05) to 0.40, 0.49, 0.53 and 0.51 g/d in the 100, 200, 400 and 800 mg/kg diets respectively. Among the herbal diets, there was no significant difference observed.

Immunological parameters

The phagocytic activity (%) showed the significant variance (P<0.05) between the control group and herbal diets fed groups in different days of challenging. Only around 29 % of the *V. harveyi* is phagocytosed in the control group. The herbals extracts increased the ingestion of the bacteria significantly (P<0.05). The highest value of 71% was observed in 400 mg/kg diet on 60th after challenge. When compared to control group, there was a significant (P< 0.05) increase in the serum bactericidal activity was observed in the herbal extract incorporated diets. In the 400 mg/kg diet, the increment was higher than the control and it was nearly about 70%. Also a significant (P<0.05) increment in A-G ratio such as 4.15, 3.84 and 3.5 respectively were observed in 400, 800 and 200 mg/kg herbal diets fed groups whereas the control group attained the least value of only 1.25 (Table 4).

Discussion

An earlier report (Yii *et al*., 1997) revealed that *Vibrio* species has been identified as a causative agent of gastroenteritis in grouper (*E. coioides)*, a disease exhibited by a swollen intestine

containing transparent yellow fluid. Applying the biochemical characteristics conformation test, this strain was characterized and identified as *Vibrio harveyi*. The bacteria could be re-isolated from kidney, liver and transparent, yellow fluid of swollen intestine. In the present study, the muscle tissues of infected grouper *E. tauvina* containing 29% of the *Vibrio sp.* were isolated. Also the percentage of *Vibrio sp.* was increased to 40% in the blood. The biochemical characterization revealed that, it is *Vibrio harveyi*, cytochrome oxidase positive to vibriostatic compounds O/129, acid but no gas produced from glucose. It is interesting to note that *V. harveyi* is mostly isolated from gastroenteritis affected carnivorous fishes, such as grouper, summer flounder, red drum, etc. Presumably the intestines of the carnivorous fish, which supply protein aceous materials from the animal sources, may favour the existence, survival or infection of the bacteria (Lee *et al*., 1999). Earlier, Saeed (1995) also studied the *Vibrio harveyi* infection in *E.tauvina*. Antibiotics and disinfectants are commonly employed for disease management, although it is not advisable due to cost effectiveness, environmental hazards and the antibiotic resistance developed by many pathogens (Kruse and Soram, 1994). To overcome this problem, herbal medicine can be used as an effective antimicrobial drug. These are the ancient Indian systems of health care and longevity (Citarasu *et al*., 2002), but still rejected by the scientific community due to lack of standardization (Ponni, 2002). In the present study, the four different extracts (butanol, petroleum ether, acetone and benzene) of the five herbals (*A. marmelos, T. bellirica, S. trilobatum, A. vasica* and *L. aspera)* effectively control the pathogen *V. harveyi*. This was supported by Jinish (2002), who extracted the plants: *Terminalia bellerica, Ocimum sanctum, Daemia extensa, Andrographis paniculata, Solanum suratense, Tinospora cardifolia, Withania somnifera* and *Myristica fragrans* and tested the extracts against the gut microfloral isolates from the grouper *E. tauvina*, and found them to have strong antimicrobial activity. Navarro *et al.* (1996) showed that methanolic plant extracts from *Eucalyptus globules, Punica granatum, Artemisia mozicana* and *Bovvania arborea* posses strong in *vitro* antibacterial activity against the *Staphylococcus aureus, Escherichia coli, Pseudomonas sp*. and *Candida sp*. In the present study, *T. bellirica* and *A. marmelos*, extracted in petroleum ether exhibited good antibacterial activity (Jinish, 2002; Jagatha, 2002).

The fish juveniles fed the herbal immuno stimulant diets had a surprising survival above 90% after the 60th day challenge and 31% higher weight than the control diet fed fish. Itami *et al.* (1998) examined the potency of oral administration of peptidoglycan derived from *Bifidobacterium thermophilum* to the kuruma shrimps through the artificial pellet diet. After 95 days, the shrimps were challenged by *Vibrio penaeicidae*. The survival and other immune parameters were significantly improved compared to the control group. Citarasu *et al.* (2002) developed an *Artemia* enriched herbal diet for *Penaeus monodon* with the combination of five herbs, which significantly increased the growth and survival during stress conditions. It was also reported that, feeding with herbal extracts such as *W. somnifera, Ferula asafoetida* and *Mucuna pruriens* remarkably accelerated the spawning rate, fecundity and larval quality in hatcheries of *P. monodon* (Babu and Marian, 2001). Earlier, several herbal principles were also tested for

their growth promoting activity in aquatic animals (Citarasu *et al*., 2002; Jayaprakas and Eupharsia, 1996).

Immunological approaches to prevent fish diseases have normally involved treatment with antibiotics, chemicals or vaccination against specific pathogens (Anderson, 1992), while the use of immunostimulants is a relatively new and developing area (Robertsen, 1999). Ecoactiva, a glucon-based immunostimulant preparation was effective in snapper *Pagrus auratus*, especially in the enhancement of the macrophage superoxide anion production. In the present study, in the 400 and 800 mg/g diet fed grouper juveniles, the haemocytes engulfed the pathogens (phagocytic activity) twice more than the control group. Also the herbal diets (200,400 and 800 mg/g) significantly influenced the Albumin-Globulin ratio, three times increased the serum bactericidal activity. The presence of antimicrobials and immunomodulatory priniciples in our experimental herbals (Nadkarni, 1995; Farouk, 1983; Tiwari *et al*., 2004) induce the immune system. Sivaram *et al*. (2004) screened the methanolic extracts of 10 antibacterial herbs against *Vibrio harveyi* isolated from the infected cultured greasy grouper (*Epinephelus tauvina)* and mixed a few selected herbal extracts in the artificial diets and fed the juvenile groupers. Among the selected ten plants, *Ocimum sanctum*, *Withania somnifera* and *Myristica fragrans* showed significant improvements in immune parameters such as phagocytic activity, serum bactericidal activity, albumin–globulin (A-G) ratio and leucocrit value.

Plant extracts such as Mistletoe (*Viscum album)*, Nettle (*Urtica dioica)*, and Ginger (*Zingiber officinalis*) on rainbow trout (*Oncorhynchus mykiss)* were investigated by Dügenci (2003). In this experiment, fish fed diets containing herbals, improved the non-specific defense mechanisms, including extracellular and intracellular respiratory burst activities, phagocytosis in blood leukocytes, total plasma protein level, specific growth rate (SGR) and condition factor (CF). Earlier studies reported that fish treated with immunostimulants showed increased phagocytosis as well as respiratory burst activity (Sakai, 1999; Siwicki, 1994; Blazer *et al*., 1984). In addition to the earlier reports, Siwicki (1989) reported that oral administration of levamisole increased the phagocytic index of phagocytic cells and it was also reported that feeding yeast products (Macro Gard, *Candida utilis*, *Saccharomyces cerevisiae*) gave rise to an enhanced phagocytic activity of blood leukocytes in rainbow trout. Albumin–globulin ratio is a measurable humoral component of the non-specific defences. The reduction of A-G in aflatoxin-treated glucan-fed fish might be due to increased total serum globulin level, with significance for the protective mechanisms of fish. In addition, the immunostimulants β-1,3 glucan and aflatoxin B1 (AFB1) improved the specific immunity against *Aeromonas hydrophila* challenge on the *Labeo rohita* fish (Sahoo and Mukherjee, 2001).

The results of the present study revealed that the herbal immuno stimulants are very effective against the *Vibrio harveyi* infection in the juvenile grouper and further work should be done on isolation and characterization of the active compounds of these herbals.

Table 1: Morphological and biochemical characteristics of the *V. harveyi* isolate from the infected Grouper *E. tauvina*

Characters	Reactions
Morpholog	Curved rods

Gram's stain	–
Motility	+
Swarming	-
Luminescence	Variable
Cytochrome oxidase	+
Nitrate reduction	+
O/129 sensitivity	+
Thornley's arginine dihydrolase	-
Lysine decarboxylase	+
Ornithine decarboxylase	+
Vogus –Proskauer reaction	-
L-Arabinose	-
M-Inositol	-
D-Arabinose	+
Sucrose	Variable

Table 2: Minimum inhibitory concentration (MIC) and Minimum Bactericidal Concentration (MBC) of petroleum ether extracts of different herbal medicinal plants against bacterial pathogen *V. harveyi* isolated from infected grouper *E. tauvina*

Concentration of extract (µg)	*Herbal extract*									
	A. marmelos		*T. bellirica*		*S. trilobatum*		*A. vasica*		*L. aspera*	
	MIC	MBC	MIC	MBC	MIC	MBC	MIC	MBC	MIC	MBC
10		+ + +		+ + +		+ + +		+ ++		+ + +
20		+ + +		+ + +		+ + +		+ ++		+ + +
30		+ + +		+ + +		+ + +		+ ++		+ + +
40		+ + +		+ + +		+ + +		+ + +		+ + +
50		+ +		+ + +		+ + +		+ + +		+ + +
60		+		+ +		+ +		+ + +		+ + +
70	70 µg	+	70 µg	+	70 µg	+		+ +		+ +
80		-		-		-	80 µg	+	80 µg	+

-: No Growth; +: Minimum Growth; ++: Intermediate Growth; +++: Maximum Growth

Table 3: Growth parameters of *E. tauvina* juveniles fed with herbal immunostimulant diets after 60 days

Diet (mg/ kg)	Initial wt (g)	Final wt (g)	Average wt gain (g /day)
Control	22.5 ± 0.33	34.40 ± 0.55	0.19 [a] ± 0.01
100	22.50 ± 0.25	46.75 ± 0.33	0.40 [b] ± 0.003
200	21.75 ± 0.55	51.5 ± 1.80	0.49 [c] ± 0.021
400	22.50 ± 1.25	54.75 ± 2.5	0.53 [c] ± 0.03
800	22.50 ± 0.33	53.40 ± 1.75	0.51 [c] ± 0.031

Means with the same superscripts do not differ from each other (P<0.05)

Table 4: Phagocytic activity (%), Albumin – Globulin ratio (A - G) and Serum bactericidal activity *E. tauvina* juveniles fed herbal immunostimulant diets after 60 days

Diet (mg /kg)	Immunological Parameters		
	Phagocytic activity (%),	Albumin – Globulin ratio (A - G)	Serum bactericidal activity
Control	28.5 ± 2.25	1.25 ± 0.02	3.55 ± 0.10
100	42.00 ± 2.00	2. 90 ± 0.13	6.33 ± 0.25
200	58. 44 ± 2.05	3.5 ± 0.54	11.71 ± 0.5
400	71.05 ± 3.33	4.15 ± 0.33	14. 28 ± 0. 80
800	70.35 ± 2.85	3.84 ± 0.40	13. 52 ± 1.58

Means with the same superscripts do not differ from each other (P < 0.05)

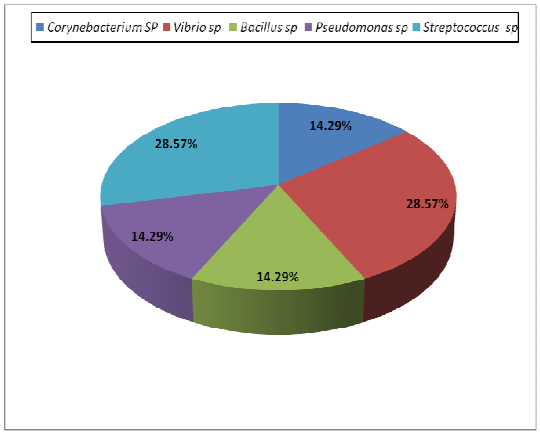

Fig. 1a: Distribution of bacterial genera isolated from the muscle tissue of infected grouper *E. tauvina*

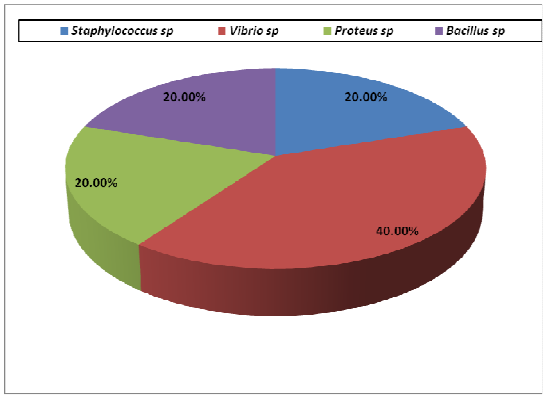

Fig. 1b: Distribution of bacterial genera isolated from the blood of infected grouper *E. tauvina*

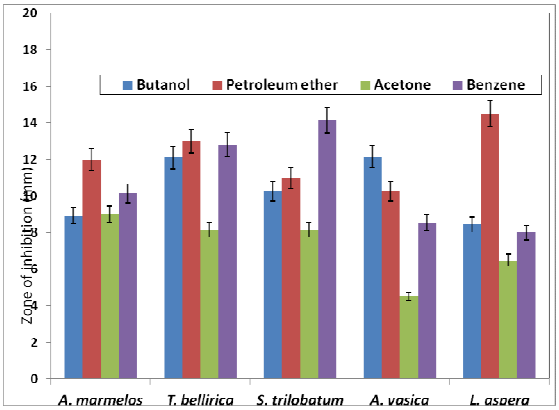

Fig. 2: Antibacterial screening of herbal extracts against *Vibrio harveyi* isolated from infected Grouper *E. tauvina*

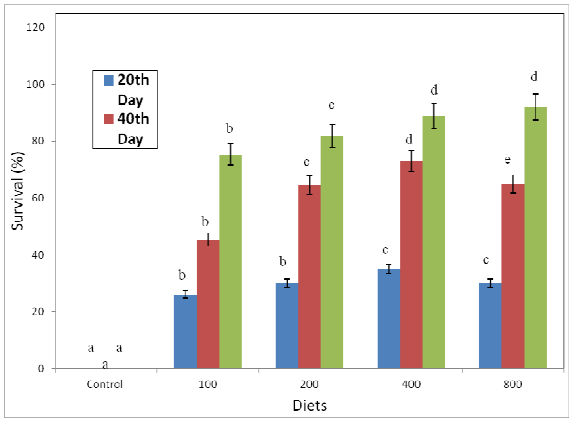

Fig. 3: Survival (%) after challenging with *V. harveyi* to the herbal immunostimulant diets fed Grouper *E .tauvina* juveniles. Same superscripts (a-e) do not differ from each other (P<0.05)

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